

Online Payments Fraud Detection: An Empirical Feature Engineering Approach on Simulated Data

Nguyen Hoang Linh^{1,*}, Le Minh Dang^{2,†}, Luu Nguyen Tan Khai^{3,†}, and Nguyen Tan Tai^{4,†}

FPT University, Ho Chi Minh Campus, Vietnam

linhnh67@fe.edu.vn¹, le5169851@gmail.com²,

tankhai15406@gmail.com³, tainguyentan0908@gmail.com^{4*}

**Main and Corresponding author.*

†These authors contributed equally to this work (Co-first authors).

Abstract

The growth of digital transactions has escalated the threat of financial fraud, necessitating effective automated detection systems. This study addresses the inherent challenge of extreme class imbalance (0.129% fraud rate) within the simulated PaySim dataset. Rather than relying on distribution-distorting resampling techniques like SMOTE, we evaluate an approach centered on domain-specific Feature Engineering, mathematically encoding auditing rules to derive balance reconciliation errors and temporal cycles. Subsequent baseline comparison across representative machine learning architectures demonstrated that LightGBM provides the most robust initial classification. To further mitigate the severe class imbalance, a systematic threshold sweep optimization was conducted. Utilizing a strict chronological train-test split, the threshold-optimized LightGBM model achieved an F1-Score of 0.998 and an ROC-AUC of 0.999. Finally, because PaySim is generated via deterministic scripts, we explicitly caution that these near-perfect metrics are intrinsically tied to the synthetic nature of the dataset; generalizing these rules-based findings to the unstructured noise of live production environments requires strict empirical re-validation.

Keywords: Fraud Detection, Imbalanced Data, Feature Engineering, LightGBM, Synthetic Data.

1 Introduction

The rapid shift toward a cashless society has expanded electronic payment services, concurrently escalating the risk of financial fraud. Because fraudulent activities inflict severe asset losses and undermine systemic trust, deploying robust, automated fraud detection systems is a critical mandate for financial institutions [1]. While prior research on the PaySim dataset heavily relies on algorithmic resampling (e.g., SMOTE) to artificially force class balance often at the risk of distorting natural financial distributions and introducing synthetic noise, a critical gap remains in exploring purely data-centric solutions. To address this, our study investigates whether meticulous, domain-inspired feature engineering alone can establish strong class separability, thereby preserving the original transaction space while maintaining highly competitive detection capabilities.

1.1 Dataset Overview and Limitations

The dataset utilized in this study is a large-scale repository simulated based on aggregated operational logs of a Mobile Money platform, commonly known as PaySim [2]. Comprising approximately 6.36 million transactions, the dataset provides an overview of monetary flows through features such as transaction type, volume, and continuous balance fluctuations of origin and destination accounts.

A prominent characteristic of this dataset is its class imbalance. Specifically, fraudulent behaviors account for approximately 0.129% (equivalent to 8,232 fraud cases). While this characteristic renders the dataset suitable for evaluating machine learning algorithms on imbalanced data, it is crucial to note that **PaySim is a synthetic/simulated dataset**. It lacks the unstructured noise (e.g., metadata, IP addresses, real-world human errors) typically found in live production environments. Therefore, insights derived from this study must be interpreted within the boundaries of this simulated environment.

Table 1: Dataset Features and Descriptions

Feature	Description & Significance
step	Represents the temporal element. One step equates to one hour of simulation.
type	The transaction type (TRANSFER, CASH_OUT, CASH_IN, PAYMENT, and DEBIT).
amount	The monetary volume of the transaction at the time of execution.
nameOrig	The identifier of the origin account (sender) initiating the transaction.
old/newbalanceOrig	The balance of the origin account prior to and subsequent to the transaction.
nameDest	The identifier of the destination account (recipient).
old/newbalanceDest	The balance of the destination account prior to and subsequent to the transaction.
isFraud	The target binary indicator denoting whether the transaction is identified as fraudulent.

1.2 Research Questions (RBL)

This research is conducted to address core data challenges (such as extreme class imbalance and overlapping distributions). The primary Research-Based Learning (RBL) questions include:

1.2.1 Which features help detect fraud earliest?

Identifying informative features is a priority. The study explores four features that provide fraud signals:

- ***errorBalanceOrig* & *errorBalanceDest*:** Logical inconsistencies within the accounting reconciliation equation (new balance minus old balance equals amount) may serve as

red flags.

- ***amount* (Transaction Volume):** Financial criminals often demonstrate a tendency to drain the entirety of the victim’s available balance in a single transaction.
- ***hour_of_day* (Temporal Cycle):** Fraudulent transactions may concentrate during off-peak hours to evade manual monitoring.

1.2.2 Which model is best at detecting fraud in imbalanced data?

Empirical analysis across selected models (including LR, RF, XGBoost, and LightGBM) is conducted to identify which algorithm provides a competitive classification performance on this imbalanced dataset.

1.2.3 What warning threshold helps reduce fraud without increasing false positives too much?

Establishing an effective decision boundary necessitates optimizing the trade-off between Precision (minimizing false alarms) and Recall (maximizing fraud detection). Instead of relying strictly on the default probability threshold (0.5), this study conducts a comprehensive threshold sweep analysis across all benchmarked models. The objective is to identify the optimal operating thresholds that successfully capture fraudulent activities while mitigating the operational penalty of excessive False Positives.

2 Related Work

The challenge of detecting fraudulent transactions in highly imbalanced financial datasets has been approached through various paradigms [1]. In the realm of open-source implementations, several practical baselines have been shared to explore the PaySim dataset. For instance, an exploratory implementation by Wakilu [9] provides a foundational analysis utilizing Exploratory Data Analysis (EDA) and baseline classifiers.

To combat extreme class imbalance, standard approaches often rely on algorithmic data-level augmentation, notably the Synthetic Minority Over-sampling Technique (SMOTE) [3], to balance the distribution prior to training. However, this dependency introduces computational overhead and risks distorting the statistical distribution of monetary flows. Recognizing these factors, our research explores a departure from resampling dependencies, pivoting toward Feature Engineeringmathematically modeling financial rulesto help advanced architectures operate directly on the raw, imbalanced distribution.

3 Methodology

3.1 Data Pre-Processing and Auxiliary Features

Initial data cleaning confirmed the absence of missing values or duplicate records. To mitigate the high cardinality of routing identifiers (*nameOrig*, *nameDest*), an auxiliary binary feature,

is_Merchant_Dest, was generated to flag merchant recipients. This dimension reduction step allows the removal of high-cardinality string IDs while retaining critical routing context.

Furthermore, the categorical transaction *type* variable was transformed via One-Hot Encoding with the *drop_first=True* parameter. This approach eliminates perfect multicollinearity by avoiding the dummy variable trap, establishing a compact numerical feature space optimized for ensemble model training.

3.2 Core Feature Engineering

While auxiliary features assist in formatting the data, constructing an informative core feature space is what primarily facilitates the algorithm in establishing a clear decision boundary. The research established two primary derived feature groups based on logical and accounting rules:

Temporal Feature: The linear sequence *step* represents simulated hours. To extract meaningful recurring patterns, the study applied a modulo mathematical transformation:

$$\text{hour_of_day} = \text{step} \pmod{24} \quad (1)$$

This logic effectively maps the continuous timeline into a standard 24-hour daily cycle, allowing the model to isolate off-peak hours.

Source & Destination Balance Errors: To evaluate the consistency of monetary flows, the research modeled fundamental auditing rules:

$$\text{errorBalanceOrig} = \text{newbalanceOrig} + \text{amount} - \text{oldbalanceOrig} \quad (2)$$

$$\text{errorBalanceDest} = \text{oldbalanceDest} + \text{amount} - \text{newbalanceDest} \quad (3)$$

Under standard conditions, these values must equal 0. Any mathematical deviation reflects a balance manipulation anomaly. It is important to acknowledge that the effectiveness of these rules-based features relies heavily on the specific data generation mechanics of the PaySim simulator.

3.3 Exploratory Data Analysis (EDA)

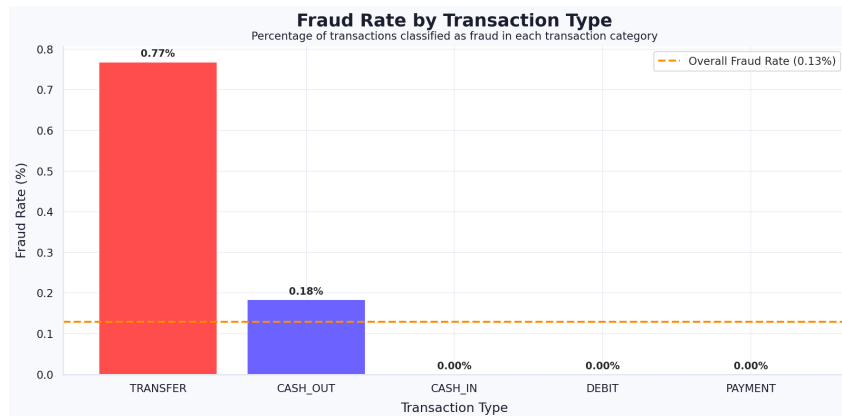


Figure 1: Distribution of fraud rates across different transaction types.

Data reveals that *CASH_OUT* and *TRANSFER* are the primary channels utilized in the simulated fraud cycle.

Temporal Feature Evaluation:

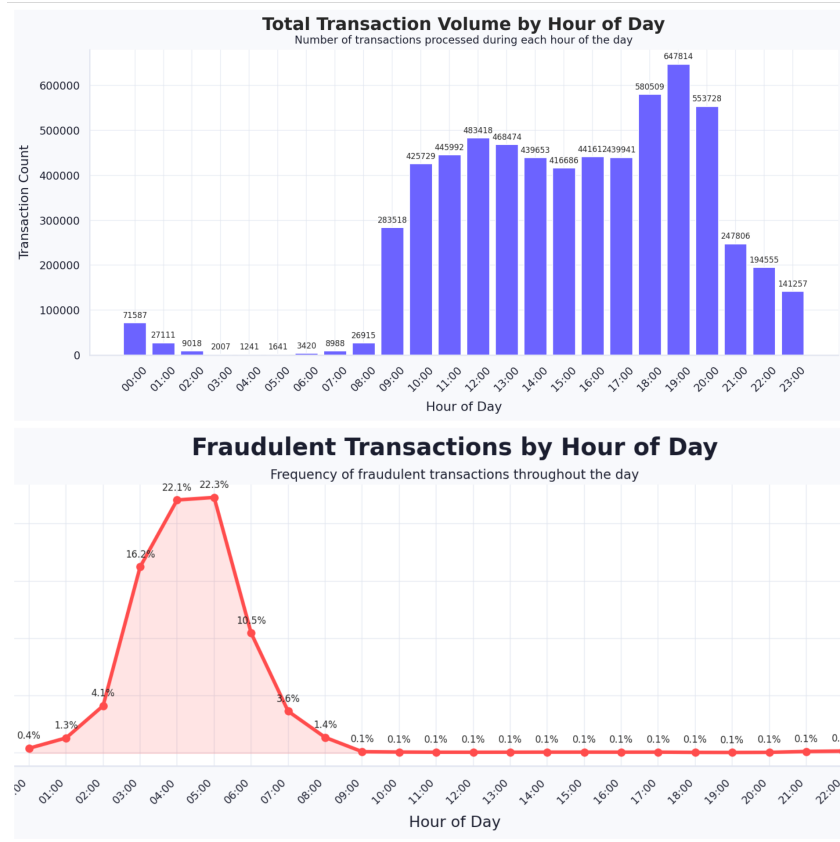


Figure 2: Total transaction volume (top) and fraudulent transaction frequency (bottom) by hour of the day.

As illustrated in Figure 2, the overall transaction volume remains low during nocturnal hours (00:00 to 07:00). Conversely, the frequency of fraudulent transactions experiences a spike during this off-peak window, peaking between 04:00 and 05:00 at over 22%. This suggests that simulated attackers may employ automated scripts to launder assets during hours of minimal legitimate traffic.

Correlation Matrix and Multicollinearity

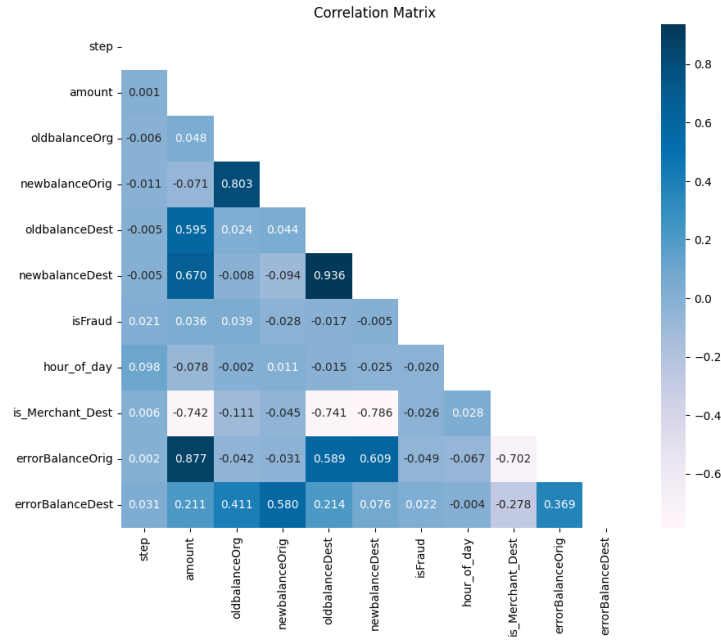


Figure 3: Correlation matrix evaluating linear relationships among features.

The correlation matrix indicates that the target variable *isFraud* approaches a 0 linear correlation with independent variables, suggesting the limitations of linear models. *errorBalanceOrig* exhibited a positive correlation (0.877) with *amount*.

Balance Error Scatter Plot Analysis:



Figure 4: Scatter plot of origin versus destination balance errors.

The scatter plot (Figure 4) reveals distinct behavioral clusters. Non-fraudulent transactions disperse widely due to natural diversities in flows. Fraudulent cases cluster along the y-axis ($\text{errorBalanceOrig} \approx 0$ but $\text{errorBalanceDest} > 0$), suggesting an attempt to maintain a valid origin balance while leaving reconciliation traces at the destination.

3.4 Algorithm Selection

Based on the dataset’s class imbalance (0.129%), this study benchmarked representative algorithms.

3.4.1 Logistic Regression (LR)

Logistic Regression [4] serves as the linear baseline. Empirical evaluation indicates that it exhibits limitations in capturing the non-linear complexity of this dataset, which translates to a relatively high False Negative rate compared to ensemble methods.

3.4.2 Random Forest (RF)

Random Forest [5] establishes non-linear decision boundaries. However, parallel construction of numerous deep trees demands high computational resources on large datasets.

3.4.3 XGBoost

XGBoost [6] successfully models non-linear patterns. However, its standard split-finding algorithm can be computationally expensive on millions of rows.

3.4.4 CatBoost

CatBoost (Categorical Boosting) [7] is a highly efficient gradient boosting framework that utilizes oblivious trees as base predictors and employs an ordered boosting scheme to prevent target leakage and reduce overfitting.

Evaluation on Dataset: CatBoost delivers highly competitive performance on this imbalanced dataset, proving the effectiveness of its unique tree-building and gradient-computation architecture.

3.4.5 LightGBM

LightGBM [8] integrates *Histogram-based Optimization* and *Leaf-wise Growth* with *Gradient-based One-Side Sampling (GOSS)*. GOSS dynamically retains rare fraud cases while subsampling legitimate cases, allowing it to partition feature spaces efficiently.

3.5 Evaluation Strategy

To ensure the integrity of the evaluation, a strict **chronological train-test split** (based on the *step* variable) was employed rather than random sampling. This temporal split prevents future information from leaking into the training phase.

Furthermore, to maintain transparency regarding the risk of **Data Leakage**, we confirm that all engineered features (including *errorBalanceOrig*, *errorBalanceDest*, and *hour_of_day*) were strictly derived from the current transaction’s state prior to knowing the target label. No features utilize future aggregations, and no variables were derived directly from pre-defined target rules.

3.6 Threshold Sweep Optimization Methodology

Relying on the default probability threshold of 0.5 often yields suboptimal classification boundaries in highly imbalanced datasets. To address Research Question 3, this study utilizes a systematic Threshold Sweep Optimization. By iteratively adjusting the decision threshold from 0.0 to 1.0 and evaluating the Precision-Recall trade-off, we identify the exact operating point that maximizes the F1-Score for each model. This post-training calibration ensures an optimal balance between maximizing fraud detection sensitivity and minimizing false alarms.

4 Results and Evaluation

This section presents a two-stage empirical evaluation of the fraud detection framework. First, a baseline benchmark compares five machine learning architectures under default settings to establish initial separability. Second, a systematic threshold optimization is conducted to align model performance with enterprise-grade requirements. In addition to standard classification metrics, Cohen’s Kappa coefficient (κ) is prioritized to evaluate operational stability. Within the financial industry, κ is often preferred over F1-score as it provides a chance-corrected measure of agreement, ensuring that the model’s alerts are structurally dependable in the face of extreme class imbalance (0.129%) [10].

Mathematically, Cohen’s Kappa is calculated as:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (4)$$

where p_o is the observed accuracy and p_e is the expected chance agreement. By penalizing this random baseline, κ robustly validates the model’s discriminative power on highly skewed data.

4.1 Benchmark Comparison

Table 2: Performance Benchmark of Selected Machine Learning Models (Default Threshold 0.5)

Model	Precision	Recall	F1-Score	PR-AUC	ROC-AUC
Logistic Regression	0.024	0.943	0.048	0.530	0.982
Random Forest	1.000	0.997	0.998	0.992	0.996
XGBoost	0.912	0.963	0.937	0.987	0.998
CatBoost	0.871	0.998	0.930	0.993	0.998
LightGBM	0.995	0.997	0.996	0.993	0.999

Table 2 indicates that Logistic Regression fails the Precision metric (2.4%). XGBoost shows strong ROC-AUC but a noticeable drop in Precision (0.912) and F1-Score (0.937). LightGBM and Random Forest demonstrate competitive performance. CatBoost also performs exceptionally well, validating the predictive efficiency of its ordered boosting architecture. At the default threshold, LightGBM maintains an excellent balance with a Recall of 0.997 and Precision of 0.995, showing an effective trade-off between classification performance and computational scalability.

4.2 Threshold Optimization and Operational Stability Analysis

Following the methodology outlined in Section 3.6, Table 3 details the empirical performance metrics before (default 0.5) and after applying calibrated thresholds. This analysis specifically monitors the evolution of Cohen’s Kappa to ensure that the optimization process enhances the model’s reliability beyond random chance.

Table 3: Impact of Threshold Sweep Optimization on Model Performance and Operational Kappa

Model	Threshold	Before (Default 0.5)				After Optimization				Δ Kappa
		Prec.	Rec.	F1	κ	Prec.	Rec.	F1	κ	
Logistic Regression	0.99	0.024	0.943	0.048	0.0461	0.323	0.623	0.425	0.4247	+0.38
Random Forest	0.18	1.000	0.997	0.998	0.9988	1.000	0.997	0.998	0.9988	0.00
XGBoost	0.90	0.912	0.963	0.937	0.9371	0.965	0.926	0.945	0.9457	+0.01
CatBoost	0.83	0.871	0.998	0.930	0.9303	1.000	0.997	0.998	0.9988	+0.07
LightGBM	0.95	0.995	0.997	0.996	0.9963	0.999	0.997	0.998	0.9984	0.00

Within the PaySim simulation environment, the empirical results reveal significant behavioral disparities among the algorithms. Logistic Regression experienced a substantial absolute increase of 0.38 in Cohen’s Kappa when the threshold was shifted to an extreme 0.99, indicating a significant reduction in chance-based false alarms. **It is important to note that this extreme calibration is specific to the synthetic distribution of PaySim and does not represent a generalizable operational standard.** Most notably, CatBoost exhibited an improvement of 0.07 in Kappa, elevating it to near-perfect agreement ($\kappa = 0.9988$) and matching the stability of the top-performing architectures. Conversely, Random Forest and LightGBM demonstrated remarkable resilience, maintaining high Kappa scores across both configurations with minimal metric deviation.

4.3 Evaluation Metrics and Visualizations

Based on the empirical evaluation of the LightGBM model on the test set of 954,393 records (containing 1,232 actual fraud cases), it is important to emphasize that these final outcomes are the direct result of combining the core feature engineering (*errorBalance*, *hour_of_day*) with the empirically optimized decision threshold (0.95). Applying this holistic configuration, the confusion matrix yields:

- True Negatives (TN) = 953,160
- False Positives (FP) = 1

- **False Negatives (FN) = 3**
- **True Positives (TP) = 1,229**

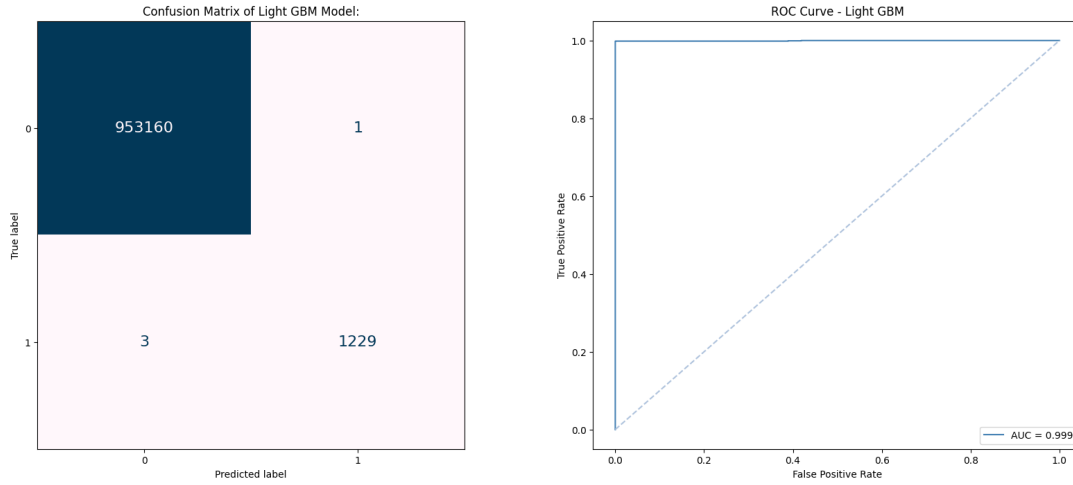


Figure 5: Confusion Matrix (left) and ROC Curve (right) for the optimized LightGBM architecture.

Derived Metrics Results:

- **Precision:** With $FP=1$ and $TP=1,229$, the Precision approaches 0.999, demonstrating the optimized threshold's success in substantially reducing false alarms.
- **Recall:** With $FN=3$ and $TP=1,229$, the Recall reaches 0.997, indicating that the engineered features effectively capture the vast majority of illicit transactions within this test set, despite the strict threshold.
- **F1-Score & ROC-AUC:** The optimized primary metric F1-Score reached 0.998, and the ROC-AUC is 0.999, which is consistent with the strong discriminative capacity of the model within this specific engineered feature space.

5 Discussion

5.1 Interpretation of Results

The empirical results indicate that the engineered reconciliation features (*errorBalanceOrig*, *errorBalanceDest*) provide highly informative signals for non-linear separation. Consequently, LightGBM ($F1 = 0.998$) outperformed linear baselines and standalone trees (e.g., Wakilu [9]), effectively handling extreme class imbalance natively without SMOTE. However, it is critical to interpret these near-perfect metrics with caution. This exceptional separability is strictly bounded by the deterministic nature of the PaySim simulator. Rather than indicating a flawless ability to detect real-world fraud, these results demonstrate that the model is efficient at mathematically isolating the rule-based anomalies specific to this synthetic environment.

5.2 The Proxy Label Dilemma and Real-World Transferability

Within the programmatic boundaries of PaySim, the engineered balance features inadvertently function as *proxy labels*. Because the simulator injects fraud via rigid, deterministic scripts (systematically emptying accounts via *TRANSFER* and *CASH_OUT*), the models mathematically exploit these simulation artifacts rather than discovering complex, hidden behavioral anomalies.

This exposes a critical divergence from real-world banking networks. In live production, applying these rigid mathematical rules would likely trigger excessive false alarms due to systemic latencies (asynchronous ledgers), benign operational noise (human errors, delayed refunds), and adaptive adversarial behaviors (e.g., smurfing). Therefore, deploying this framework in a real-world setting requires integrating broader behavioral profiles and multimodal risk indicators.

5.3 Limitations and Practical Implications

The primary limitation of this research is its reliance on synthetic data, necessitating empirical re-validation of the *errorBalance* features on live production datasets. Nevertheless, the findings demonstrate the practical value of histogram-based gradient boosting architectures. Their high computational efficiency and low false-positive rates suggest a robust foundational framework for scanning high-throughput financial transactions, provided the feature space is continuously adapted to real-world operational realities.

6 Conclusion and Future Work

This research explored a methodology for detecting fraudulent transactions within an imbalanced simulated financial dataset (0.129% fraud rate). The findings indicate that the synergy between domain-specific Feature Engineering (accounting balance errors and temporal cycles) and gradient boosting architectures (LightGBM) yields highly competitive metrics, achieving an optimized F1-Score of 0.998 without resorting to resampling techniques.

Ultimately, the results suggest that domain-specific feature engineering acts as the primary driver of predictive performance within this simulated setting, while rigorous model selection serves as the critical final step to optimally exploit the engineered feature space.

Future Work: Future research should focus on applying this methodology to real-world, non-synthetic datasets to validate the robustness of the engineered features and consider implementing Graph Neural Networks (GNNs) to capture the topological complexities of money laundering rings.

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